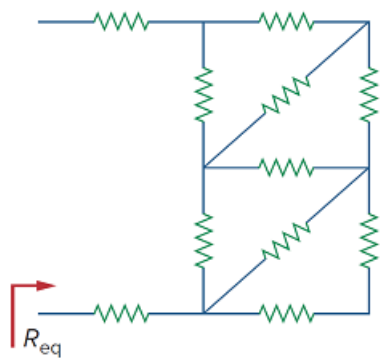


Problem

For the circuit shown in Fig. 2.116, find the equivalent resistance. All resistors are 3Ω .

Figure 2.116:



Step-by-step solution

Step 1 of 8

Draw the circuit diagram as shown in Figure 1.

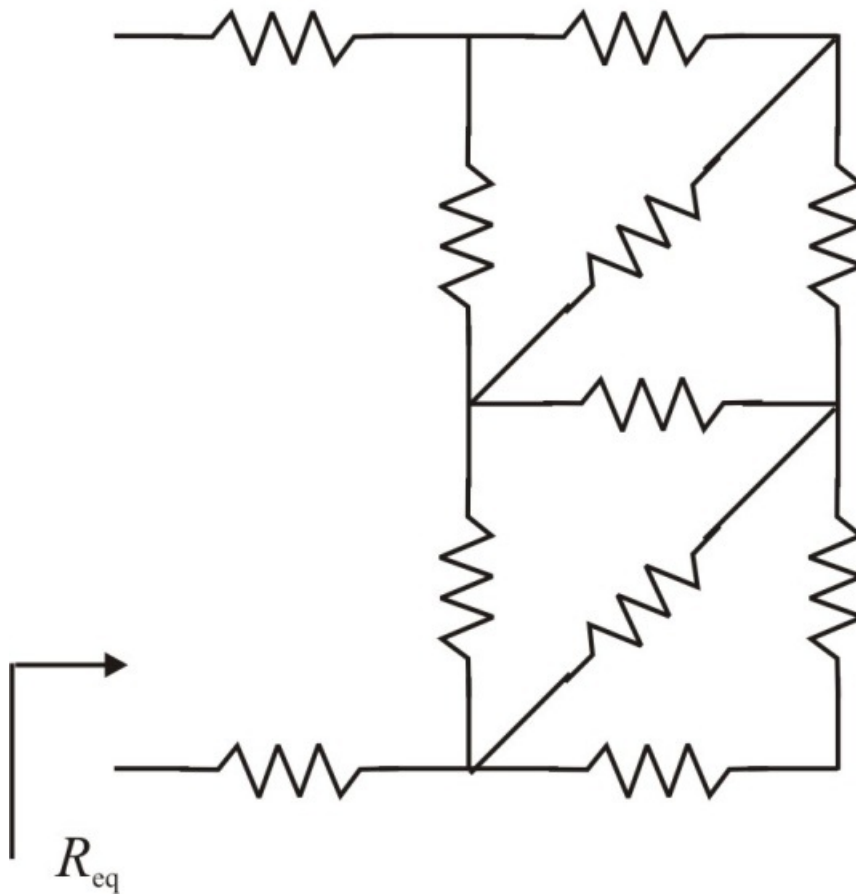


Figure 1

Step 2 of 8

The value of all resistors $= 3\Omega$

Convert Y network comprising the 3Ω , 3Ω and 3Ω resistors to Δ network, by selecting

$$R_1 = 3\Omega$$

$$R_2 = 3\Omega$$

$$R_3 = 3\Omega$$

Then, the resistance R_a can be calculated as,

$$\begin{aligned} R_a &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \\ &= \frac{3 \times 3 + 3 \times 3 + 3 \times 3}{3} \\ &= 9\Omega \end{aligned}$$

The resistance R_b can be calculated as,

$$\begin{aligned} R_b &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \\ &= \frac{3 \times 3 + 3 \times 3 + 3 \times 3}{3} \\ &= 9\Omega \end{aligned}$$

The resistance R_c can be calculated as,

$$\begin{aligned} R_c &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \\ &= \frac{3 \times 3 + 3 \times 3 + 3 \times 3}{3} \\ &= 9\Omega \end{aligned}$$

Then the circuit in Figure 1 is reduced to that in Figure 2.

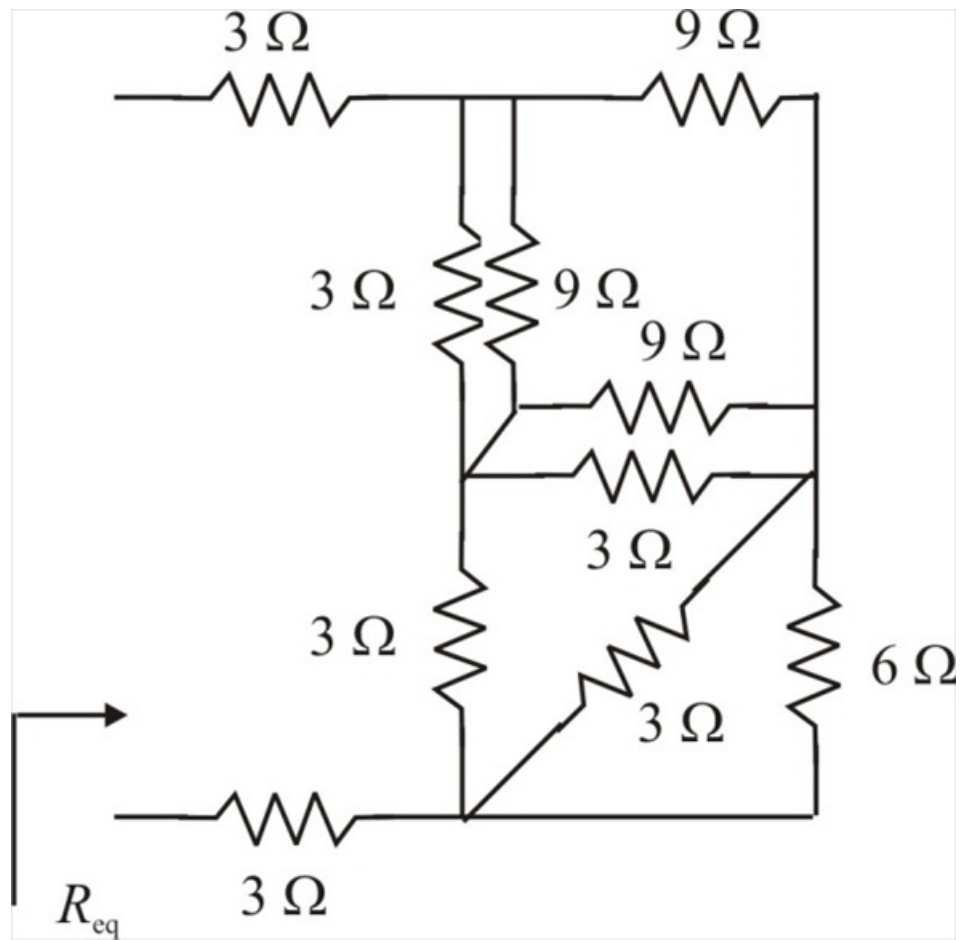


Figure 2

The 9 Ω and 3 Ω resistors are in parallel. Their combined resistance is:

$$9\Omega \parallel 3\Omega = \frac{(9)(3)}{9+3} \\ = 2.25\Omega$$

The 9 Ω and 3 Ω resistors are in parallel. Their combined resistance is:

$$9\Omega \parallel 3\Omega = \frac{(9)(3)}{9+3} \\ = 2.25\Omega$$

The 3 Ω and 6 Ω resistors are in parallel. Their combined resistance is:

$$3\Omega \parallel 6\Omega = \frac{(3)(6)}{3+6} \\ = 2\Omega$$

Then the circuit in Figure 2 is reduced to that in Figure 3.

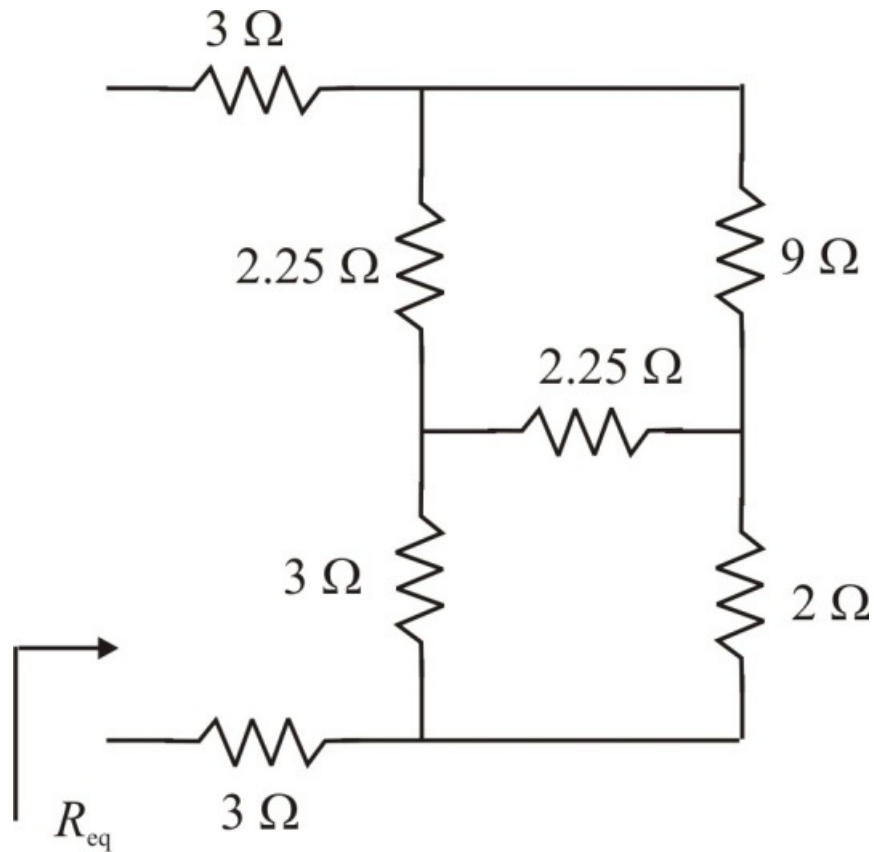


Figure 3

Then convert Y network comprising the 3 Ω , 2.25 Ω and 2.25 Ω resistors to Δ network, by selecting

$$R_1 = 3 \Omega$$

$$R_2 = 2.25 \Omega$$

$$R_3 = 2.25 \Omega$$

Then, the resistance R_a can be calculated as:

$$\begin{aligned} R_a &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \\ &= \frac{(3)(2.25) + (2.25)(2.25) + (2.25)(3)}{3} \\ &= 6.188 \Omega \end{aligned}$$

The resistance R_b can be calculated as:

$$\begin{aligned} R_b &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \\ &= \frac{(3)(2.25) + (2.25)(2.25) + (2.25)(3)}{2.25} \\ &= 8.25 \Omega \end{aligned}$$

The resistance R_c can be calculated as:

$$\begin{aligned} R_c &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \\ &= \frac{(3)(2.25) + (2.25)(2.25) + (2.25)(3)}{2.25} \\ &= 8.25 \Omega \end{aligned}$$

Step 6 of 8

Then the circuit in Figure 3 is reduced to that in Figure 4.

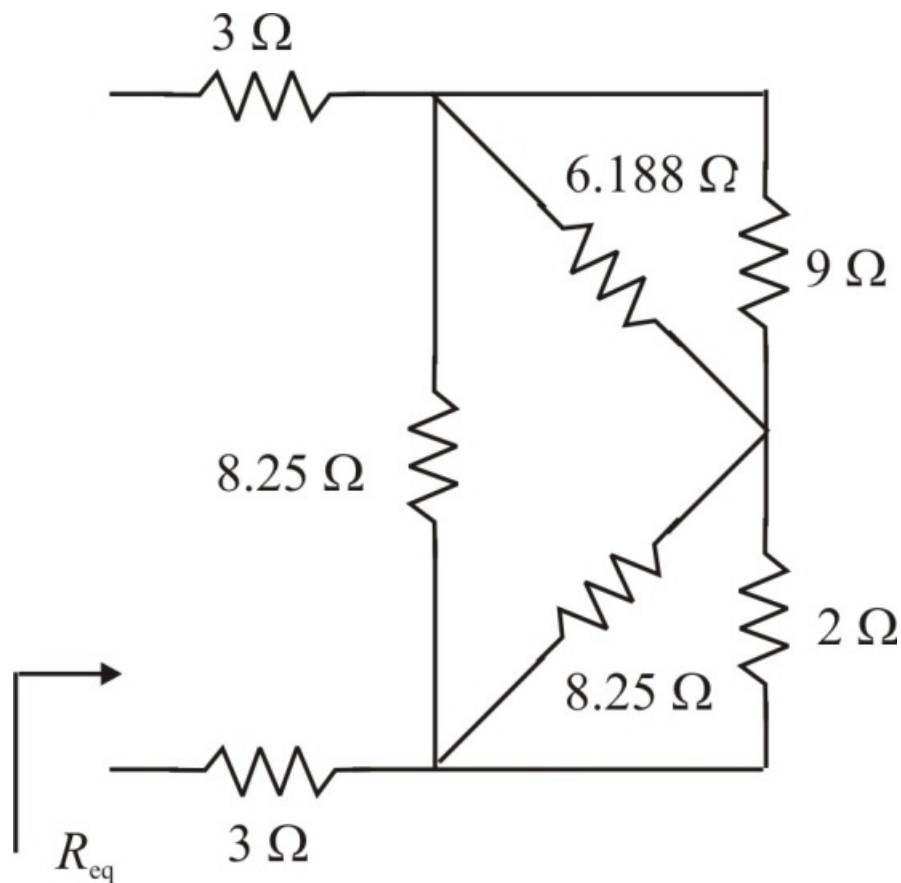


Figure 4

Step 7 of 8

The $6.188\ \Omega$ and $9\ \Omega$ resistors are in parallel. Their combined resistance is,

$$6.188\ \Omega \parallel 9\ \Omega = \frac{(6.188)(9)}{6.188+9} \\ = 3.666\ \Omega$$

The $8.25\ \Omega$ and $2\ \Omega$ resistors are in parallel. Their combined resistance is,

$$8.25\ \Omega \parallel 2\ \Omega = \frac{(8.25)(2)}{8.25+2} \\ = 1.61\ \Omega$$

This $1.61\ \Omega$ and $3.666\ \Omega$ resistors are in series. Their combined resistance is,

$$1.61\ \Omega + 3.666\ \Omega = 5.276\ \Omega$$

Then the circuit in Figure 4 is reduced to that in Figure 5.

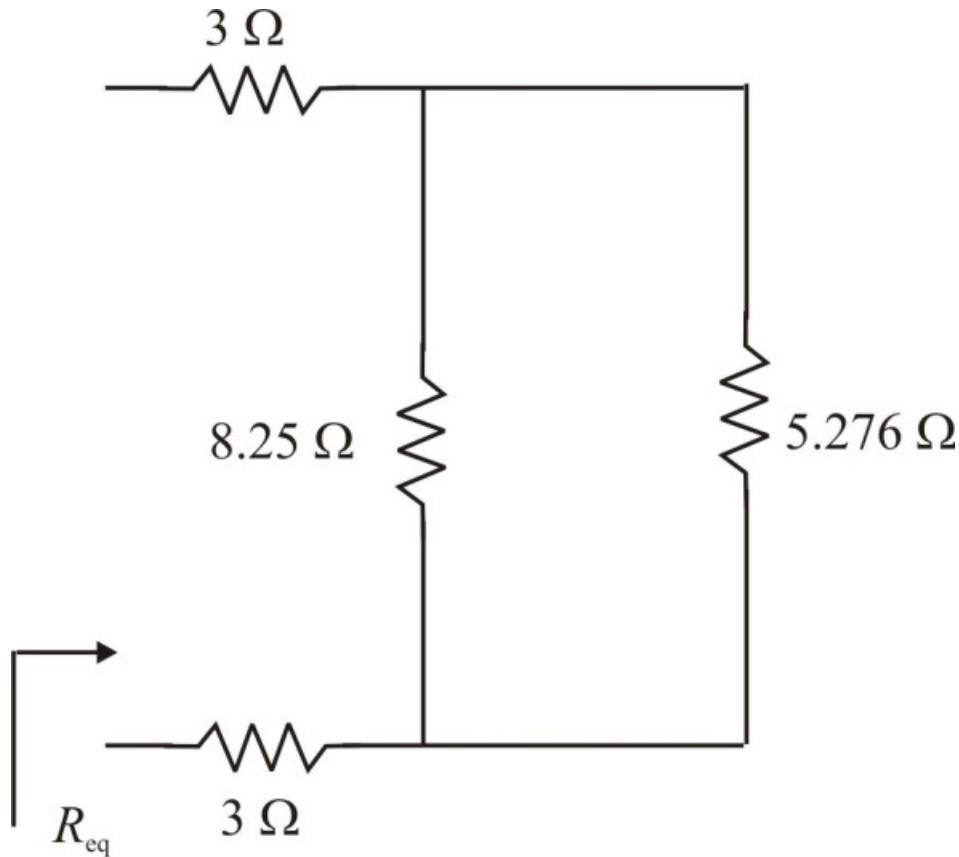


Figure 5

Step 8 of 8

The $8.25\ \Omega$ and $5.276\ \Omega$ resistors are in parallel. Their combined resistance is:

$$8.25\ \Omega \parallel 5.276\ \Omega = \frac{8.25 \times 5.276}{8.25 + 5.276} \\ = 3.218\ \Omega$$

This $3.218\ \Omega$, $3\ \Omega$ and $3\ \Omega$ resistors are in series. Their combined resistance is:

$$3.218\ \Omega + 3\ \Omega + 3\ \Omega = 9.218\ \Omega$$

Therefore the equivalent resistance R_{eq} is $9.218\ \Omega$.